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SQL Basics Quick Guide

A beginner-friendly walkthrough of core SQL concepts

1 What is SQL?

SQL (Structured Query Language) is used to store, retrieve, manage and manipulate data in relational databases.

Database
A structured collection of data.

DBMS
Software used to create, manage and interact with databases.

Relational Database
Stores data in tables made up of rows and columns.

SQL vs MySQL
SQL — language to communicate with databases. **MySQL** — a DBMS that uses SQL.

2 How a Query Works

Application → SQL Query → Database → Result

3 Sample Table & Core Commands

`students`

id	name	age	city
1	Rahul	21	Jaipur
2	Priya	22	Sikar
3	Aman	20	Ajmer

`SELECT`
Retrieve data

`WHERE`
Filter records

`ORDER BY`
Sort results

`LIMIT`
Restrict result count

```sql
SELECT * FROM students;

SELECT name, city
FROM students
WHERE age > 20;

SELECT *
FROM students
ORDER BY age DESC
LIMIT 2;
```

Quick Tip: SQL keywords are generally written in uppercase for readability, but SQL is usually not case-sensitive.

02 / 02 — QUERIES & PRACTICE

Essential SQL Queries

Filtering, sorting, aggregation & joins

1 AND / OR

```
SELECT *
FROM students
WHERE age > 20 AND city = 'Sikar';
```

2 COUNT() & AVG()

```
SELECT COUNT(*) FROM students;

SELECT AVG(age) FROM students;
```

3 GROUP BY

Groups rows with the same value so aggregate functions can be applied.

```
SELECT city, COUNT(*)
FROM students
GROUP BY city;
```

4 JOINs

Used to combine related data from multiple tables.

`students`

id	name	course_id
1	Rahul	101
2	Priya	102

`courses`

course_id	course_name
101	Java
102	React

```
SELECT students.name, courses.course_name
FROM students
INNER JOIN courses
ON students.course_id = courses.course_id;
```

INNER JOIN returns records that have matching values in both tables.

5 Mini Practice — Try These 5 Queries

1. Display all students.
2. Find students whose age is greater than 20.
3. Sort students by name.
4. Count the total number of students.
5. Display the average age of students.

Want to go deeper into SQL?
Explore the complete developer resources for detailed SQL notes, examples, interview preparation and practice material.

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